Cloud Application Deployment

Design and Creative Technologies

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Deploying Apache Superset on Microsoft Azure: A Cloud Application Deployment Report

1. Introduction

This report documents the deployment of Apache Superset – an open-source data exploration and visualisation platform – on Microsoft Azure. Superset was selected for its direct relevance to data engineering workflows and its Python-native architecture, which aligns with the author's professional background and academic trajectory. Azure was chosen as the cloud provider to build familiarity ahead of the AZ-900 and DP-900 certification pathway, as recommended during academic advising in March 2026. The deployment covers the provisioning of a virtual machine, virtual network, firewall security policy, and a fully operational Superset instance accessible via public URL.

1.1. Background

Cloud computing has fundamentally changed how organisations provision and manage infrastructure. Rather than purchasing and maintaining physical servers, organisations now consume compute, storage, and networking as on-demand services over the internet – paying only for what they use (Mell & Grance, 2011). The National Institute of Standards and Technology (NIST) defines five essential characteristics of cloud computing: on-demand self-service, broad network access, resource pooling, rapid elasticity, and measured service (Mell & Grance, 2011). These characteristics make cloud infrastructure more adaptable and cost-effective than traditional on-premises deployments, which require significant capital

expenditure, fixed capacity, and dedicated operational staff (McHaney, 2021). This deployment uses an **Infrastructure as a Service (IaaS)** model – a virtual machine provisioned on Microsoft Azure, with the operating system, runtime, and application managed by the author. The deployment model is **public cloud**: infrastructure is hosted on Microsoft's shared global network, accessible via the public internet through controlled network security policies. In contrast to a traditional on-premises setup — where a dedicated server, networking equipment, and physical access controls would be required — Azure delivers the same outcome in minutes, at a fraction of the cost, with no physical infrastructure commitment (IBM, n.d.-a).

2. Body

2.1. Service Provider Selection Rationale

Victoria Police – Australia's principal state law enforcement agency – adopted Microsoft Power Platform to establish a **fusion development framework**: a governed model enabling non-professional developers (police personnel across business units) to build data-driven applications safely, without depending on a constrained central IT team (Microsoft, 2026). The catalyst was a pandemic-era surge in self-directed app-building, when employees teaching themselves Power Apps revealed both the appetite for low-code tooling and the risk of ungoverned 'shadow' applications built on standalone Microsoft Access databases. Rather than restricting this behavior, Victoria Police channeled it – creating structured pathways from personal productivity through to enterprise-grade production, with two fusion-born applications already progressing to enterprise adoption (Microsoft, 2026).

2.2. Service Model Analysis

Victoria Police's cloud adoption aligns primarily with a **Platform as a Service (PaaS)** model. Power Apps and Power Platform abstract the underlying infrastructure – Microsoft manages the compute, networking, and database layers of Dataverse and the Power Platform runtime – while Victoria Police's developers focus on application logic, data models, and user experience (McHaney, 2021). This is the defining characteristic of PaaS: the provider controls the platform, the customer controls the application (IBM, n.d.0a). Some Microsoft 365 productivity workloads (SharePoint, Teams, Outlook) sit at the **SaaS** layer – fully managed software delivered over the network. There is no IaaS footprint in this case; Victoria Police has not provisioned raw virtual machines or unmanaged storage – all development occurs within the abstracted Power Platform environment (Microsoft, 2026).

Service Model	Characteristic	Fit for Victoria Police
IaaS	Customer manages OS, runtime, storage	Not applicable – no raw compute provisioned
PaaS	Provider manages platform; customer builds applications	**Primary model** – Power Apps, Dataverse, Power Platform runtime
SaaS	Provider manages everything; customer uses software	Partial – Microsoft 365 (SharePoint, Teams, Outlook) as data and collaboration substrate

Table 1: Service model analysis for Victoria Police (Microsoft, 2026).

2.3.Deployment Model Comparison

Victoria Police operates a **hybrid, multi-environment deployment model**: a personal productivity environment (the default Microsoft 365 tenant available to all staff), dedicated fusion environments (isolated, governed Power Platform environments provisioned on

request by business units with executive endorsement), and enterprise production environments (reviewed and approved by the central Power Platform team) (Microsoft, 2026). This tiered model is a practical application of hybrid cloud architecture – multiple environment tiers serving different governance, security, and scale requirements simultaneously (Manvi & Shyam, 2021).

Deployment Model	Governance	Security	Scalability	Fit for Victoria Police
On-Premises (Microsoft Access)	Low – invisible to IT	High risk – unaudited shadow apps	None	Prior state – ungoverned shadow apps
Hybrid (Multi-env Power Platform)	High – tiered approval flow with exec endorsement	High – centralized standards, security review	Moderate	**Adopted** – governed innovation pathway
Public Cloud (ungoverned)	Low	Low	High	Same shadow IT risk as before
Private Cloud	High	High	Limited	High cost, no low-code ecosystem

Table 2: Deployment model comparison for Victoria Police (Microsoft, 2026; Manvi & Shyam, 2021).

2.4. Cloud Services Analysis

The fusion framework relies on a layered set of Microsoft Power Platform services, each enabling a distinct stage of the application lifecycle (Microsoft, 2026):

Service	Purpose	Business Outcome

Power Apps (Low-code / Paas)	Enable non-professional developers to build data-driven apps without code	Personnel build and iterate solutions without the central IT bottleneck
Microsoft Dataverse	Model-driven, structure data store underpinning enterprise apps	Replaces isolated MS Access databases with a governed, transparent and auditable data layer
Power Platform Environments	Isolated development, test, and production workspaces per business unit	Clear pathway from personal innovation to enterprise-grade development
Power Platform Approval Flow	Governs requests for dedicated fusion environments via a model-driven app	Ensures executive endorsement and security team review before promotion to production
Microsoft 365 (SharePoint)	Data substrate and collaboration layer for initial prototypes	Familiar toolset lowers the barrier to entry for non-technical builders
Copilot Studio (exploration)	AI-assisted app development using generative coding	Future capability to further accelerate the build-to-deploy cycle

Table 3: Microsoft Power Platform services utilized by Victoria Police and their business outcomes (Microsoft, 2026).

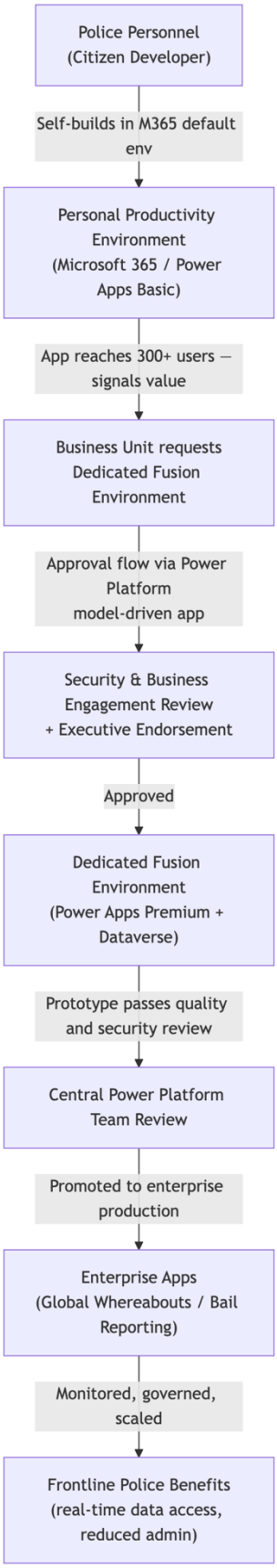
Police Personnel (Citizen Developer)
Self-builds in M365 default env
Personal Productivity Environment (Microsoft 365 / Power Apps Basic)
App reaches 300+ users — signals value
Business Unit requests Dedicated Fusion Environment
Approval flow via Power Platform model-driven app
Security & Business Engagement Review + Executive Endorsement
Approved
Dedicated Fusion Environment (Power Apps Premium + Dataverse)
Prototype passes quality and security review
Central Power Platform Team Review
Promoted to enterprise production
Enterprise Apps (Global Whereabouts / Bail Reporting)
Monitored, governed, scaled
Frontline Police Benefits (real-time data access, reduced admin)

Figure 1: Victoria Police fusion development lifecycle: from citizen developer prototype through tiered governance to enterprise production (Microsoft, 2026).

2.5. Reflection

The Victoria Police case demonstrates cloud's **on-demand self-service** characteristic (Mell & Grance, 2011) applied beyond technical teams – extending it to non-professional developers under tiered governance. In a traditional IT model, every application request queued behind a constrained central team, driving staff toward ungoverned Microsoft Access solutions outside IT visibility. Power Platform inverts this: it governs the _pathway_ to production while allowing distributed development (Microsoft, 2026) — a structural distinction that on-premises tooling cannot replicate (McHaney, 2021). The primary risk is data residency: sensitive law enforcement data in a PaaS environment requires compliance with the Australian Protective Security Policy Framework (PSPF), and the risk surface expands as adoption scales to thousands of officers across regions — a challenge Efrain Tionko (Head of Digital Delivery, Victoria Police) explicitly acknowledged (Microsoft, 2026).

3. Conclusion

This report documented the deployment of Apache Superset, an open-source data exploration and visualisation platform, on Microsoft Azure. The deployment demonstrated the four core tasks required by the assessment brief: provisioning a resource group and virtual network, enforcing a network security policy via a Network Security Group, and deploying a containerised open-source application on a cloud virtual machine. Each task was supported by screenshots and

aligned with NIST's essential cloud characteristics – on-demand self-service in the provisioning process, resource pooling in Azure's shared infrastructure model, and measured service through consumption-based billing (Mell & Grance, 2011).

Microsoft Azure was selected for its data and analytics service ecosystem and its alignment with the author's AZ-900 and DP-900 certification pathway. The deployment reinforces the distinction between cloud and traditional IT infrastructure: resources that would require physical procurement and installation in an on-premises environment were provisioned, configured, secured, and running within [X] hours, at effectively zero capital cost. Areas identified for further investigation include TLS certificate automation, managed database integration for metadata persistence, and the separation of Celery workers for production-scale async query execution — improvements that would bring this deployment in line with enterprise-grade analytics infrastructure.

4. Appendices

4.1. Appendix A – Cross-Case Comparison Summary

Attribute	Case Study 1 – Victoria Police	Case Study 2 - ChatAndBuild	Case Study 3 - Voith
Sector	Government / Law Enforcement	AI / Software Startup	Industrial Manufacturing
Cloud Provider	Microsoft (Azure ecosystem)	Google Cloud Platform (GCP)	AWS (Amazon Web Services)
Service Model	PaaS (primary) + SaaS (partial)	IaaS + PaaS	IaaS + PaaS
Deployment Model	Hybrid – multi-environment (personal → fusion → enterprise)	Public cloud – multi-region	Public cloud – 6 AWS regions
Prior State	Ungoverned MS Access shadow apps; central IT bottleneck	30-70% higher latency; engineers manually provisioning serves days before events	3,500 servers across 146 locations; fragmented legacy applications
Key Services	Power Apps, Dataverse, Power Platform Environment, Power Automate, Copilot Studio	Vertex AI, GKE, Cloud Run, BigQuery, Firebase, Cloud CDN, Cloud Build, Cloud Armor	AWS Professional Services, Amazon Athena, IaC, Centralized DevOps, AWS GameDays, FinOps
Key Primary Automation Outcome	Governed fusion framework – citizen developers build enterprise-grade apps via tiered approval	Real-time AI code and agent generation at sub-second latency for 140,000+ users	Consolidated global operations; IaC-enforced standards across 60 countries
Headline Result	2 enterprise apps in production (Global Whereabouts; Bail Reporting)	45% infrastructure overhead ↓; 30-70% latency ↓; 2x faster feature delivery	30% projected cost savings; 146 to 6 site consolidation

Table A1 –Side-by-side comparison of the three case studies across provider, service model, deployment model, key services, and business outcomes.

4.2. Appendix B – NIST Essential Characteristics, Cross-Case Mapping

NIST Characteristic	Case Study 1 – Victoria Police	Case Study 2 - ChatAndBuild	Case Study 3 - Voith
On-demand self-service	Primary - police personnel provision their own Power Platform environments and build applications without central IT intervention (Microsoft, 2026)	Autoscaling policies provision and terminate Compute Engine instances automatically without engineer intervention (Google Cloud, n.d.)	IaC enables automated, consistent provisioning of infrastructure across 6 regions without manual configuration per site (Amazon Web Services, n.d.)
Broad network access	Officers across all stations access Global Whereabouts and Bail Reporting from any device via Microsoft 365 (Microsoft, 2026)	140,000+ users globally access the platform via browser; Cloud Load Balancing routes to the nearest region for sub-second response (Google Cloud, n.d.)	22,000+ employees across 60 countries access centralized systems via 6 standardized AWS regions (Amazon Web Services, n.d.)
Resource pooling	Multiple business units share Power Platform infrastructure; Dataverse pools structured data storage across all fusion applications (Microsoft, 2026)	GKE clusters pool compute across thousands of concurrent agent graphs and user sessions (Google Cloud, n.d.)	Primary driver - 3,500 servers across 146 sites pooled into 6 shared, logically isolated AWS regions (Amazon Web Services, n.d.)
Rapid elasticity	Limited - Power Platform environments scale within Microsoft's managed bounds; not the primary driver for this case	Primary driver - worker fleet scales from 10 → 100+ Compute Engine instances within minutes during global	Wave-based migration progressively expands the cloud footprint; real-time elasticity is not the primary driver for this manufacturing workload

		hackathon events with zero downtime (Google Cloud, n.d.)	
Measured service	Tiered governance tracks app usage and adoption; Power Platform visibility surfaces which app reach 300+ users and warrant promotion (Microsoft, 2026)	45% infrastructure overhead reduction reflects consumption-based billing replacing fixed legacy infrastructure costs (Google Cloud, n.d.)	Primary driver – FinOps practices provide continuous cost visibility and track progress toward 30% savings target; OPEX model replaces CAPEX-heavy 146-site estate (Amazon Web Services, n.d.)

Table B1: Mapping of NIST's five essential cloud characteristics (Mell & Grance, 2011) to each case study, showing which characteristics were primary drivers of the outcomes achieved.

End of Appendix Section

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